Ahmedabad University

CSE 519: HUMAN-COMPUTER INTERFACE

Part A

Mid-semester Exam

Submitted To : Prof Anurag Lakhlani

Name: Shlok Shelat
Enrollment number: AU2240025
Programme: BTech in Computer Science and Engineering

Link To the Video:
https://drive.google.com/file/d/1u0Wqr4GT0E0PT44uSwwB_nJT7SVWg3rL/view?usp=sharing

Introduction

This report details the Human-Computer Interaction (HCI) design and paper-based low-fidelity prototyping for an imaginary "Time Machine". The primary objective of this project is to create an intuitive interface that allows a user to travel to any location on Earth at any time in the past or future.

The design process focuses on translating complex temporal and geographical logic into a user-friendly sequence of screens, incorporating essential data entries such as user identification, destination coordinates, temporal settings, stay duration, and return protocols.

Design Philosophy

Drawing inspiration from the 22nd-century gadgets of the *Doraemon* universe, this prototype leverages strong **metaphors**(such as the "Anywhere Door") and **physical affordances** (such as the "Time Cloth" dials) to bridge the gap between the system and the user's mental model.

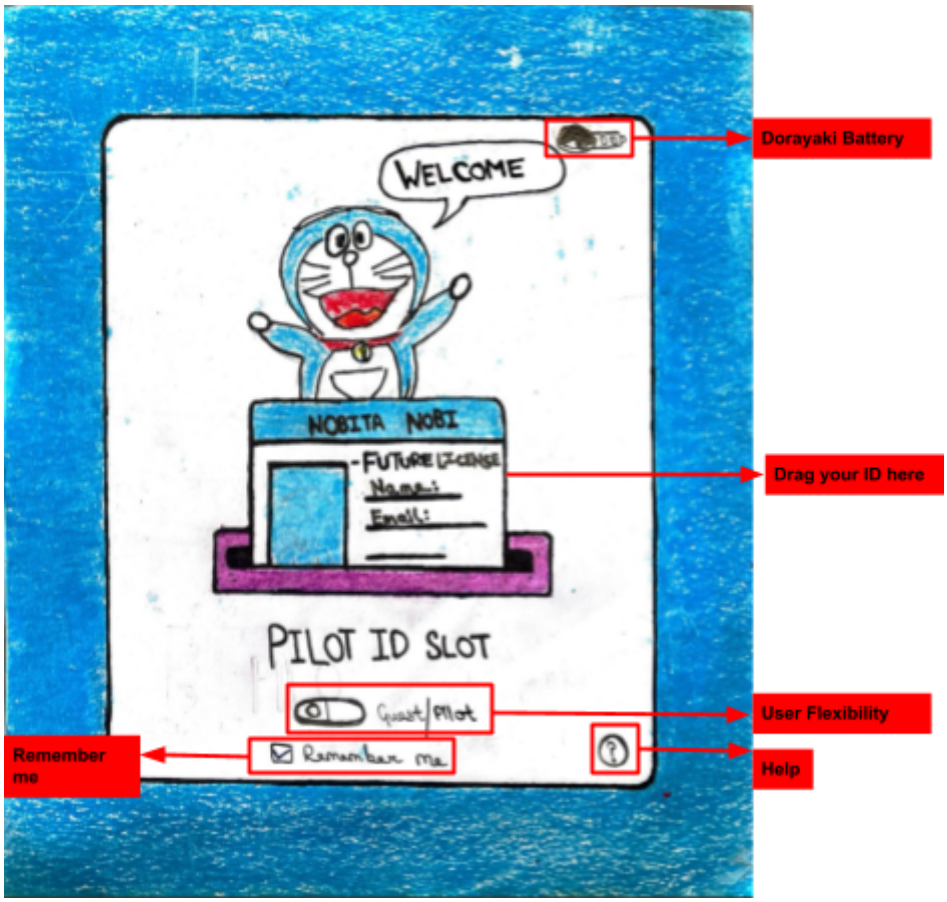
Core HCI Objectives

The interface is constructed following the "Principles of Good HCI Design" learned in class, specifically focusing on:

- **Visibility of System Status:** Keeping the user informed through feedback and status indicators.
- **Error Prevention and Handling:** Implementing at least two distinct error screens—a critical constraint for "Time Paradoxes" and a constructive validation message for missing data—to illustrate high-quality error management.
- **Universal Usability:** Ensuring that interaction styles, such as direct manipulation and tangible inputs, accommodate varying levels of user expertise.

By utilizing a paper-based design, this project demonstrates how low-fidelity prototyping can effectively communicate interaction sequences and system responses before moving into high-fidelity development.

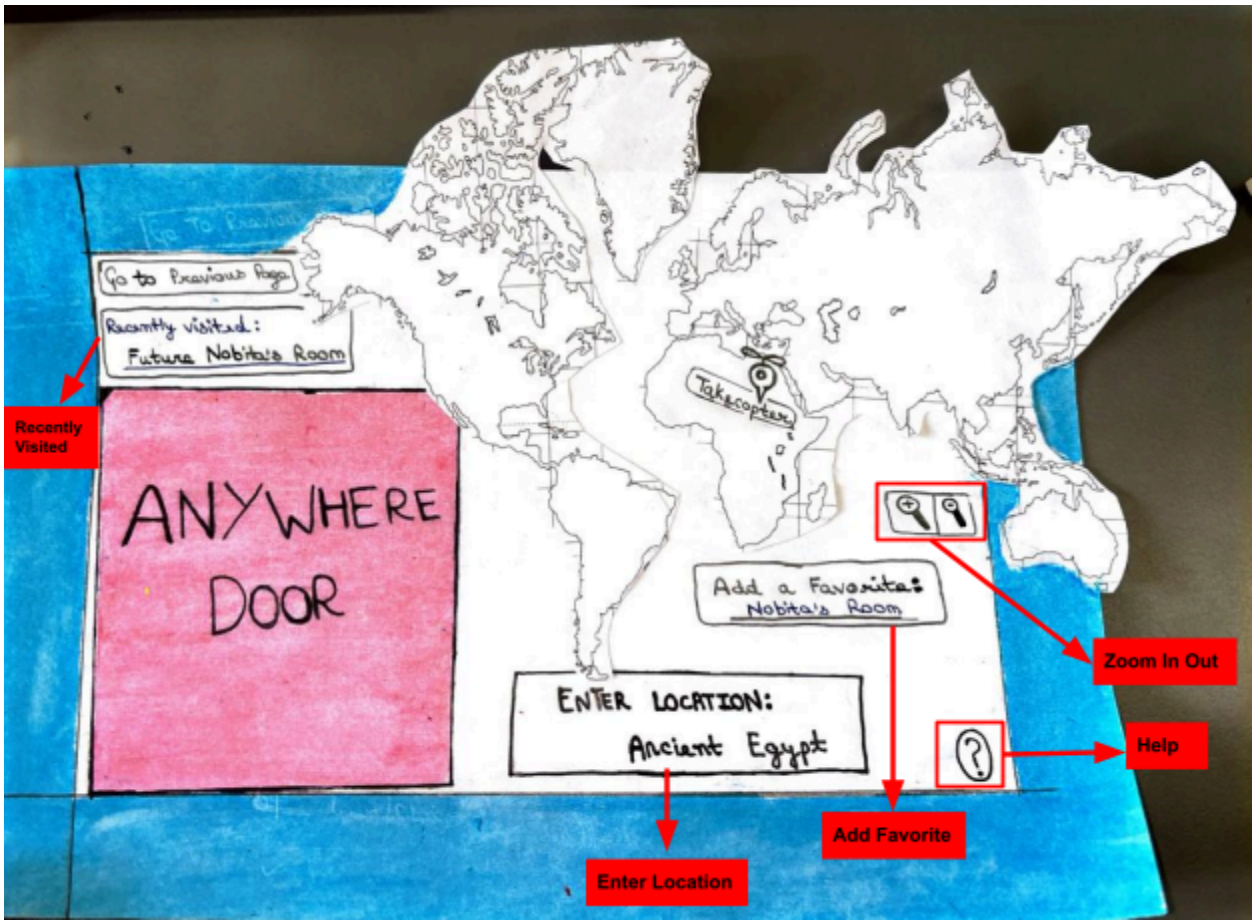
Screen 1: User Authentication (Pilot ID Interface)



Description:

- **Function:** Secure user identification and system activation.
- **Interaction Style:** Tangible User Interface (TUI) combined with physical toggles.
- **HCI Principles Implemented:** This screen utilizes a TUI where inserting a physical card provides immediate **Visual Feedback** via the pop-up Doraemon. The top-right indicators ensure constant **Visibility of System Status**(battery/signal). The "Guest/Pilot" toggle supports **Flexibility and Efficiency of Use** for different user types, while the "?" icon offers on-demand **Help and Documentation**.

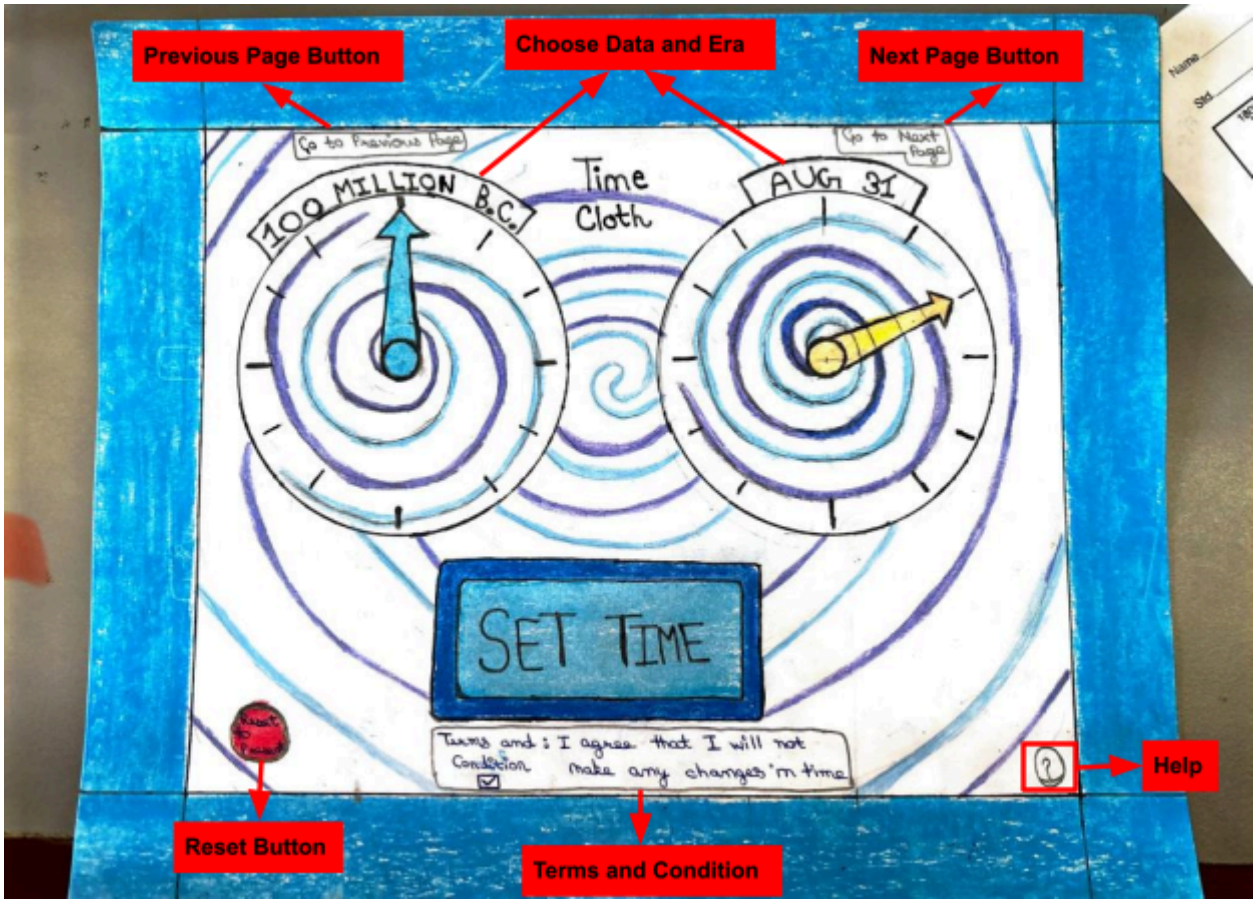
Screen 2: Destination Selection (The Anywhere Door)



Description:

- **Function:** Selection of geographical coordinates for travel.
- **Interaction Style:** Direct Manipulation (pin placement) and Menu Selection.
- **HCI Principles Implemented:** The design leverages the strong **Metaphor** of the "Anywhere Door," aligning with the user's **Mental Model** of instant travel. The "Favorites" sidebar improves **Efficiency of Use** by allowing expert users to bypass the map search. Zoom controls provide necessary **User Control and Freedom** to adjust the map view.

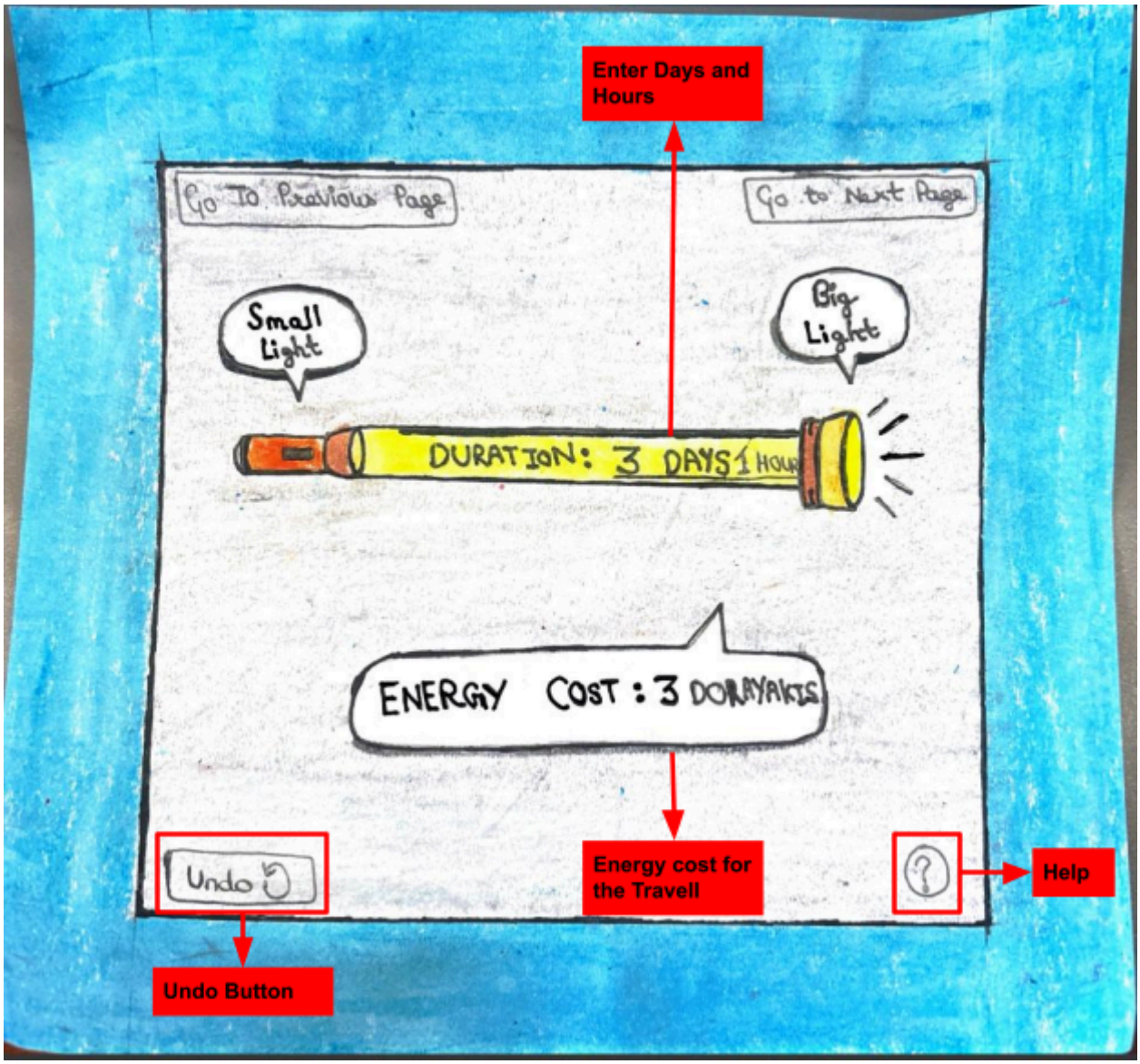
Screen 3: Temporal Navigation (Time Cloth Dials)



Description:

- **Function:** Setting the specific target date and era.
- **Interaction Style:** Rotary Control (skeuomorphic dials).
- **HCI Principles Implemented:** The rotary dials offer excellent physical **Affordance** for spinning. Colored zones on the dials act as **Error Prevention** constraints, visually warning against dangerous time periods (e.g., "Big Bang"). The prominent "RESET TO TODAY" button acts as an **Emergency Exit**, supporting **User Control and Freedom** to quickly recover from complex inputs.

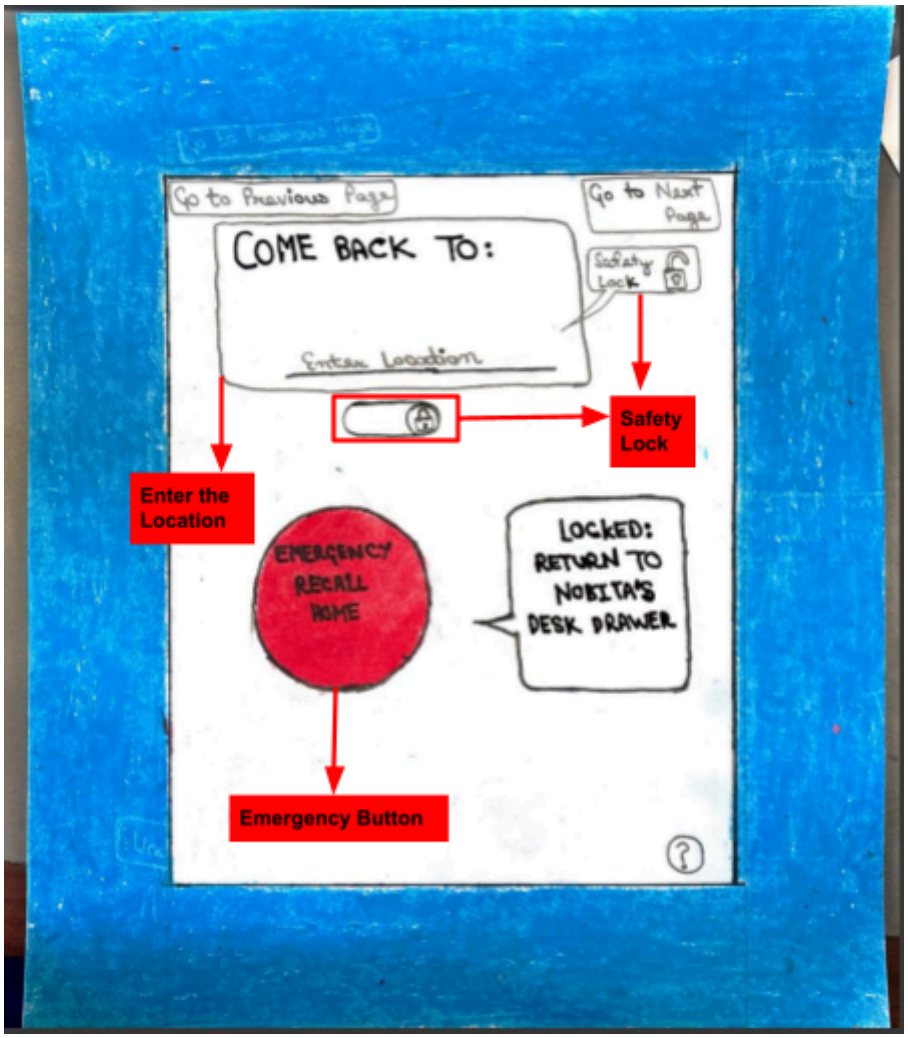
Screen 4: Duration Control (Light Slider)



Description:

- **Function:** Defining the length of stay at the destination.
- **Interaction Style:** Slider Control.
- **HCI Principles Implemented:** This screen uses **Conceptual Mapping** where sliding towards the "Big Light" intuitively corresponds to increasing duration. The "Energy Cost" display provides immediate **Consequential Feedback** on the user's choice. The dedicated "UNDO" button supports **Recoverability**, allowing users to immediately reverse motor slips without restarting the task.

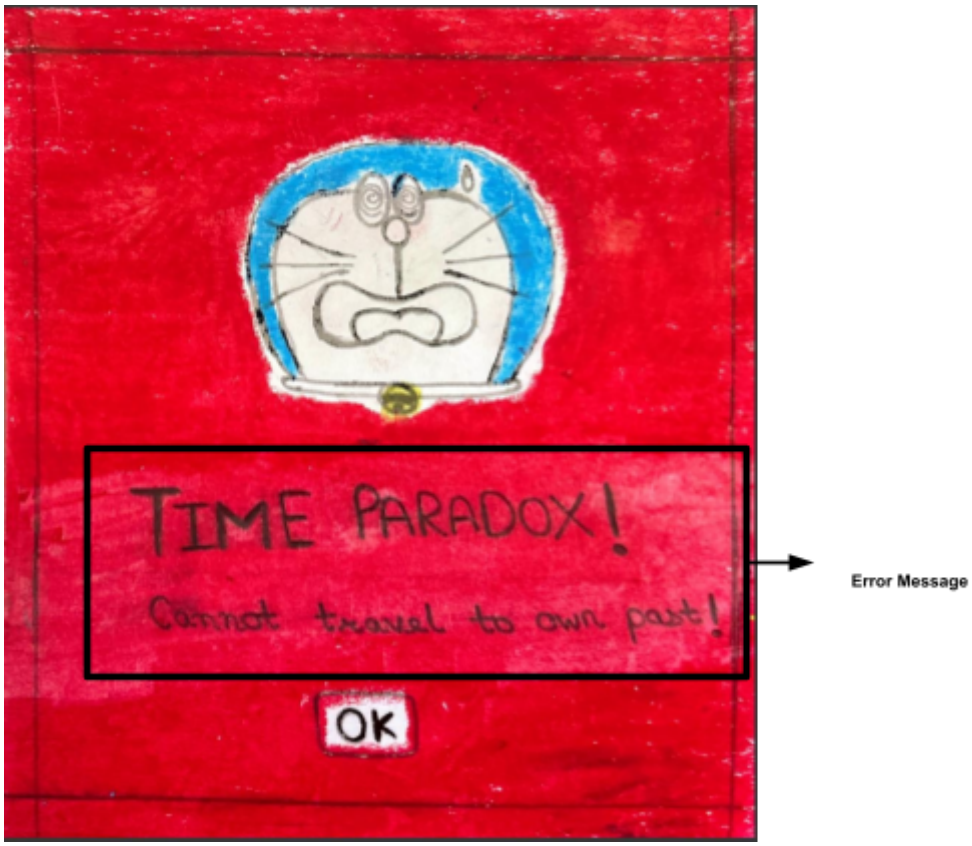
Screen 5: Return Protocol (Safety Lock)

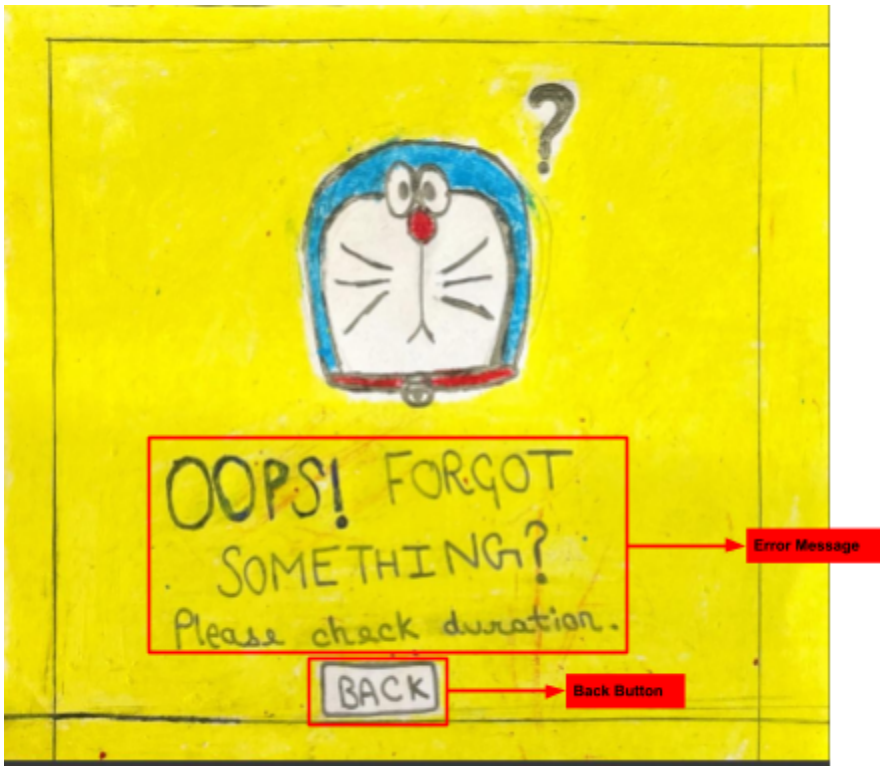


Description:

- **Function:** Establishing and confirming the safe return vector.
- **Interaction Style:** Physical Button with Safety Cover.
- **HCI Principles Implemented:** To prevent catastrophic accidental activation, a physical **Forcing Function** (the hinged safety lock) must be engaged before the button is accessible. The return location uses a **Smart Default**("Nobita's Desk Drawer") to ensure user safety, reducing cognitive load during a critical phase.

Examples of Error Handling (Warning Cards)





Description:

- **Function:** System alerts for invalid or dangerous inputs.
- **HCI Principles Implemented:** The Red Card acts as a hard **constraint**, preventing fatal errors by explicitly stating the logic ("Cannot travel to own past!") in plain language. The Yellow Card provides polite, **constructive Feedback** for missing data, offering an easy recovery path via the "Back" button.